Password Expiry Notifier & Locked Account Notifier

I will use mailtrap.io for its email sandbox that will catch emails when the admin (me) needs to be notified of password expiry or locked account events.

1. (Password Expiry Notifier) After the onboarding lab, I know it's not set up, but I want to make sure the AD users I created have email addresses set:

Blank as expected

```
PS C:\WINDOWS\system32> Get-ADUser -Filter * -SearchBase "OU=New Hires,OU=LAB Users,DC=LAB,DC=local" -Properties EmailAddress | Select-Object Name, EmailAddress

Name            EmailAddress
----            ------------
Jane Doe
Mike Johnson
Sarah Williams
Lisa Brown
Gabriel Marrero
```

2. Setting up the email addresses for all those accounts now using a PowerShell script:

setupEmailAddresses.ps1

```
$Users = Get-ADUser -Filter * -SearchBase "OU=New Hires,OU=LAB Users,DC=LAB,DC=local"

ForEach ($User in $Users) {

    $Email = "$($User.SamAccountName)@LAB.local"
    Set-ADUser -Identity $User -EmailAddress $Email
    Write-Host "Set email for $($User.SamAccountName) to $Email"
}
```

```
PS C:\Users\Administrator> C:\Scripts\Notifications\setupEmailAddresses.ps1
Set email for jdoe to jdoe@LAB.local
Set email for mjohnson to mjohnson@LAB.local
Set email for swilliams to swilliams@LAB.local
Set email for lbrown to lbrown@LAB.local
Set email for gmarrero to gmarrero@LAB.local
```

Confirmation:

```
PS C:\WINDOWS\system32> Get-ADUser -Filter * -SearchBase "OU=New Hires,OU=LAB Users,DC=LAB,DC=local" -Properties EmailAddress | Select-Object Name, EmailAddress

Name            EmailAddress
----            ------------
Jane Doe        jdoe@LAB.local
Mike Johnson    mjohnson@LAB.local
Sarah Williams  swilliams@LAB.local
Lisa Brown      lbrown@LAB.local
Gabriel Marrero gmarrero@LAB.local
```

3. In this lab environment, passwords expire after 42 days. Also, the only user who set a password was gmarrero (the intern) when I did the DHCPServer lab. I will need to force a password set on all of them.

setupEmailAddresses.ps1 | forcePasswordSet.ps1

```
$Users = Get-ADUser -Filter * -SearchBase "OU=New Hires,OU=LAB Users,DC=LAB,DC=local"

ForEach ($User in $Users) {

    Set-ADAccountPassword -Identity $User -Reset -NewPassword (ConvertTo-SecureString "Password1!" -AsPlainText -Force)
    Write-Host "Password set for $($User.SamAccountName)"
}
```

```
PS C:\Users\Administrator> C:\Scripts\Notifications\forcePasswordSet.ps1
Password set for jdoe
Password set for mjohnson
Password set for swilliams
Password set for lbrown
Password set for gmarrero
```

Confirmation:

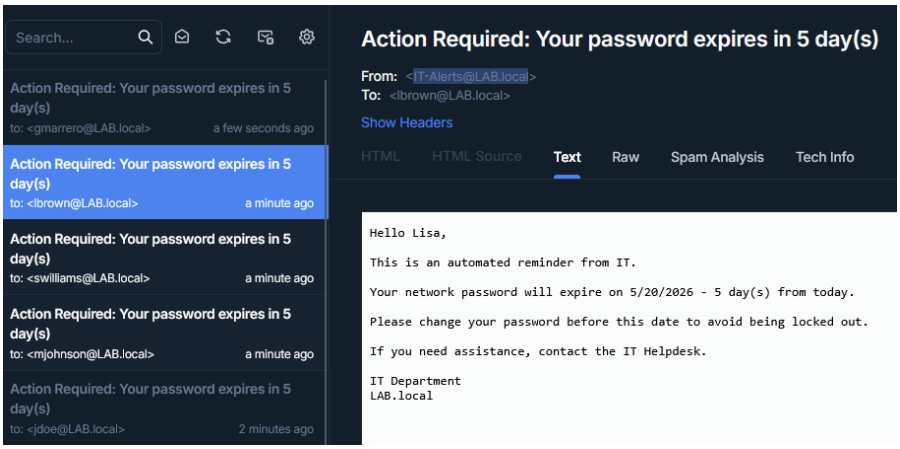
```
PS C:\WINDOWS\system32> Get-ADUser -Filter * -SearchBase "OU=New Hires,OU=LAB Users,DC=LAB,DC=local" -Properties PasswordLastSet | Select-Object Name, PasswordLastSet

Name            PasswordLastSet
----            ---------------
Jane Doe        4/20/2026 10:08:16 AM
Mike Johnson    4/20/2026 10:08:16 AM
Sarah Williams  4/20/2026 10:08:16 AM
Lisa Brown      4/20/2026 10:08:16 AM
Gabriel Marrero 4/20/2026 10:08:16 AM
```

4. Building the Password Expiry Notifier script using PowerShell which will scan AD for users whose passwords expire within a threshold (5 days to 0) and send them an email notification.

Script is in separate file

```
PS C:\Users\Administrator> C:\Scripts\Notifications\passwordExpiryNotifier.ps1
2026-04-23 09:24:02 - Starting password expiry check, 5 user(s) found.
2026-04-23 09:24:02 - EXPIRING - jdoe password expires in 5 day(s) on 5/20/2026.
2026-04-23 09:24:03 - EMAIL SENT - Notified jdoe at jdoe@LAB.local.
2026-04-23 09:24:18 - EXPIRING - mjohnson password expires in 5 day(s) on 5/20/2026.
2026-04-23 09:24:18 - EMAIL SENT - Notified mjohnson at mjohnson@LAB.local.
2026-04-23 09:24:33 - EXPIRING - swilliams password expires in 5 day(s) on 5/20/2026.
2026-04-23 09:24:33 - EMAIL SENT - Notified swilliams at swilliams@LAB.local.
2026-04-23 09:24:48 - EXPIRING - lbrown password expires in 5 day(s) on 5/20/2026.
2026-04-23 09:24:48 - EMAIL SENT - Notified lbrown at lbrown@LAB.local.
2026-04-23 09:25:03 - EXPIRING - gmarrero password expires in 5 day(s) on 5/20/2026.
2026-04-23 09:25:03 - EMAIL SENT - Notified gmarrero at gmarrero@LAB.local.
2026-04-23 09:25:18 - Password expiry check complete.
```

Confirmation:



1. (Locked Account Notifier) – First I will simulate a locked account on one of the users I have in the New Hires OU. The simple method is to fail the login attempts to trigger a lockout.

simulateAccLockout.ps1

```
$Password = ConvertTo-SecureString "NotThePassword1!" -AsPlainText -Force
$Cred     = New-Object System.Management.Automation.PSCredential("LAB\gmarrero", $Password)

1..10 | ForEach-Object {

    Try {
        Start-Process cmd -Credential $Cred -WindowStyle Hidden -ErrorAction Stop
    } Catch {}
}
```

```
PS C:\Users\Administrator> C:\Scripts\Notifications\AccountLockoutNotifier\simulateAccLockout.ps1

PS C:\Users\Administrator> get-aduser -Identity gmarrero -Properties LockedOut | Select-Object Name,LockedOut

Name            LockedOut
----            ---------
Gabriel Marrero      True
```

```
PS C:\WINDOWS\system32> get-aduser -Identity jdoe -Properties LockedOut | Select-Object Name,LockedOut

Name     LockedOut
----     ---------
Jane Doe      True
```

2. Building the AccountLockoutNotifier script in PowerShell to alert the admin what accounts are locked.

```
PS C:\Users\Administrator> C:\Scripts\Notifications\AccountLockoutNotifier\AccountLockoutNotifier.ps1
2026-04-24 09:08:03 - Starting locked account check. 2 locked account(s) found.
2026-04-24 09:08:03 - Locked - jdoe
2026-04-24 09:08:03 - Locked - gmarrero
2026-04-24 09:08:03 - EMAIL SENT - Admin alerted at admin@LAB.local for 2 locked account(s).
```

Email confirmation:

IT Alert: 2 Locked Account(s) Detected

From: <IT-Alerts@LAB.local>
To: <admin@LAB.local>

Show Headers

HTML HTML Source **Text** Raw Spam Analysis Tech Info

```
IT Admin,

The following Active Directory accounts are currently locked out:

 - Jane Doe (jdoe)
 - Gabriel Marrero (gmarrero)

Please investigate and unlock accounts as appropriate.

This is an automated alert from the LAB.local domain.

IT Automation
LAB.local
```